



1 Basic Java for Programming

2 Printing Patterns

🔗 Easy → Medium → Hard

Easy Patterns

Medium Patterns

Hard Patterns

💧 30+ Pattern Problems Covered

3 Functions

Call by Value

Call by Reference

💧 10+ Function-Based Problems

4 Basic Programming Problems

Digits, Numbers, Math Logic

Fibonacci, GCD, Prime, Divisors, etc.

💧 15+ Core Logic Problems

5 Arrays

Searching & Traversal

Rotations & Reversal

Missing / Special Elements

Subarray Basics

💧 25+ Array Problems

6 Time & Space Complexity

Big-O / Big-Ω / Big-Θ

Loop & Recursive TC

Space Analysis

💧 10+ Conceptual Problems

7 Sorting Algorithms

Bubble, Selection, Insertion

Counting & Cyclic Sort

Missing Number Variants

💧 20+ Sorting Problems

8 Binary Search

Classic BS

Rotated Arrays

Answer-based BS

Hard Interview Problems

💧 35+ Binary Search Problems

9 Two Pointers & Popular Algorithms

Pair / Triplet / Quadruple

Kadane's Algorithm

Dutch National Flag

Moore's Voting

💧 45+ Problems

10 Prefix / Suffix Pattern

Product Except Self

Rain Water Trapping

Interval Based Problems

💧 20+ Problems

1 1 2D Arrays / Matrix

Spiral, Wave, Rotate

Matrix Binary Search

Median / Peak

💧 20+ Matrix Problems

1 2 Strings

Palindrome

Conversions

Pattern Matching

Large Number Handling

🔥 20+ String Problems

1 3 Hashing (HashMap / HashSet)

Frequency

Subarray Sum

Longest Subarray

🔥 15+ Hashing Problems

1 4 Sliding Window

Fixed & Variable Window

Substring / Subarray Patterns

🔥 25+ Sliding Window Problems

1 5 Recursion & Backtracking

Basics to Advanced

Subsets / Permutations

N-Queens / Sudoku

🔥 40+ Recursion & Backtracking Problems

1 6 Dynamic Programming

🔗 Divided into 10 Powerful Patterns

Fibonacci & Knapsack

LCS Variants

LIS Variants

Matrix DP

Game Theory

Tree DP

MCM

🔥 70+ Dynamic Programming Problems

1 7 OOPs (Programming Oriented)

Classes & Objects

Inheritance

Polymorphism

☒ Sufficient OOPs for DSA & Interviews

1 **8** Stack & Queue

Monotonic Stack

Parenthesis Problems

Expression Evaluation

 35+ Stack & Queue Problems

1 **9** Linked List

Single & Doubly LL

Cycle, Reverse, LRU Cache

 25+ Linked List Problems

2 **0** Trees & BST

Traversals

Views

Diameter, LCA

BST Operations

 35+ Tree Problems

2 **1** Heap / Priority Queue

Top K Pattern

Two Heaps

Scheduling Problems

 30+ Heap Problems

22 Graphs

🔗 BFS, DFS, Dijkstra, Union Find, Topo

💧 50+ Graph Problems

23 Trie

Prefix Based Searching

XOR Problems

💧 15+ Trie Problems

24 Bit Manipulation

XOR Patterns

Power Problems

Gray Code

💧 25+ Bit Manipulation Problems

25 Segment Trees

Range Queries

Lazy Propagation

💧 10+ Segment Tree Problems